

Chapter 1: States of matter

Solids, liquids & gases



Solids



- Fixed volume, Fixed shape, high density.
- Atoms vibrate in place
- Closely packed together & regular pattern



Liquids



- Fixed volume & take shape of container
- Less dense than solids, more dense than gas
- Particles move and slide past each other.

Gases



- not fixed volume, take shape of container
- Lowest density, a lot of space between particles so gases are easily compressed.
- Random particles & particles move quickly.
- Particles collide with each other & the sides of the container (this is how pressure is created).

State changes

Melting

- Solid $\rightarrow$ Liquid
- heat energy $\rightarrow$ kinetic energy and melting only occurs at specific temp known as melting point.

Boiling

- Liquid $\rightarrow$ Gas
- heat needed to form bubbles of gas below the surface of liquid, allowing liquid to escape from surface.
- occurs at boiling point (b.p.).

Freezing

- Liquid $\rightarrow$ Solid
- reverse of melting, occurs at same temperature
- requires lot of temp decrease

Evaporation

- liquid $\rightarrow$ gas at any temperature.
- evaporation happens at the surface only
- high energy particles escape surface at any temperature.
- the larger the surface area, the more quickly a liquid evaporates.

Condensation

- gas $\rightarrow$ liquid at range of temperatures.
- when a gas is cooled, particles lose energy & when they bump into each other they don't have enough energy to bounce away, instead they group together and form a liquid.

Kinetic theory

When substance is heated, particles absorb thermal energy which is converted to kinetic energy.

Heating solid causes particles to vibrate more, & as temperature increases they vibrate to the point solid's structure breaks & it melts.

With more heating, the liquid expands as well until particles at surface gain enough energy to escape & the liquid boils when bp reach. These changes are shown on a heating curve & reverse is a cooling curve, these curves show how changes in temperature effects changes of state.

Pressure & temperature in gases

The more the temperature increases, the volume of gas also increases. And the density decreases if volume increases.

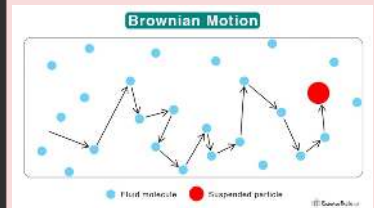
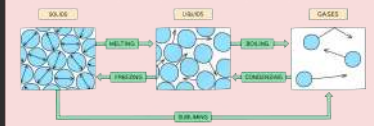
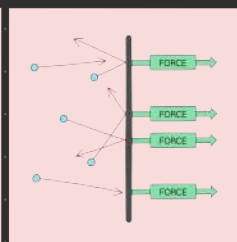
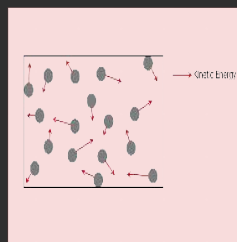
If you have gas stored in container & squeezed, the pressure increases as you decrease volume.

Gases & kinetic theory

Gaseous particles are in constant & random motion.

Pressure that gas creates inside a closed container is produced by gas hitting the walls of container.

An increase in kinetic energy increases if temperature increases. With increase in kinetic energy, particles move faster and collide with walls more frequently. Decrease in volume = increase in pressure.

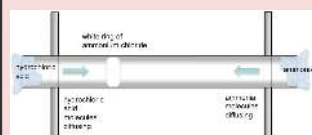
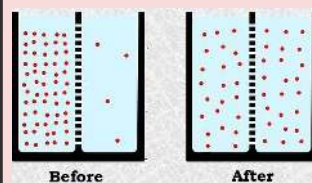


Diffusion

Diffusion is the movement of particles from area of high concentration to an area of low concentration and eventually concentration of particles will be even as they spread out to occupy all the space. Diffusion happens on its own & doesn't require energy but it happens faster at higher temperatures. Diffusion occurs faster in gases compared to liquids. At same temperature, gases don't diffuse at same rate. This is due to difference in molecular masses, lighter particles travel faster, therefore they diffuse faster.

This can be demonstrated in the reaction between ammonia (NH_3) and hydrogen chloride gas (HCl) inside glass tube. Where the two gases meet, white smoke of ammonium chloride (NH_4Cl) is formed.

However, they don't meet in the middle of the tube instead they meet closer to the end where hydrogen chloride ($M_r = 36.5$) is, this is because ammonia ($M_r = 17$)'s molecules are lighter so they travel faster than HCl .



Extra notes:

Chapter 2: Atoms, Elements and Compounds

elements, compounds & mixtures



elements

Substance made up of atoms that contain the same proton number & can't be split into something simpler.

compounds

Substance made of two or more elements chemically combined. there's unlimited compounds. they can't be separated.

Mixtures

Mixture of two or more substances (elements and/or compounds) that aren't chemically combined. they can be separated by physical methods.



Oxygen

Name: Oxygen
Atomic Number: 8
Symbol: O
Atomic Weight: (15.99903, 15.99907)
Electron Configuration: $1s^2 2s^2 2p^4$

Electron
Nucleus
Orbit

Visit <https://www.ck12.org/element/oxygen/>

Electronic Configuration

Using diagrams called electron shell diagrams called the electronic config.

Electron Shell diagrams

electrons orbit the nucleus in shells (AKA energy levels), further from nucleus there is more energy per shell.

electrons fill the shell closest to the nucleus first.

first shell holds 2 electrons, 2nd shell holds 8 electrons and for the first 20 elements, the third shell & above can also hold 8 but this is a simplified version.

the outermost shell is called the valence shell.

the arrangement of electrons can also be explained by writing the number of electrons in each shell and separating by commas.

this is the electronic configuration.

atomic structure & periodic table

- all substances are made of tiny particles of matter called atoms.
- each atom has subatomic particles: Protons, neutrons, electrons.
- the protons and neutrons are located at the centre of the atom, which is called the nucleus.
- electrons move very fast around nucleus in orbital paths called shells.
- mass of electrons is negligible, hence the mass of an atom is contained within the nucleus where protons & neutrons are located.
- relative atomic mass unit is $1/12^{\text{th}}$ the mass of a carbon-12 atom
- all elements are measured relative to the mass of a carbon-12 atom so relative atomic mass has no units.

Relative mass & charge:

- Proton 1 Positive 1+
- Neutron 1 no charge / neutral 0
- Electron $\frac{1}{1840}$ negative 1-

Proton number

atomic number / Proton number is the number of protons in the nucleus.

the symbol for atomic number is Z.

number of electrons present in a neutral atom & determine position of the element on the periodic table.

Mass Number

total number of protons & neutrons, the symbol is A

Nucleon (mass) - Proton number gives neutron number.

Periodic table shows element together with their atomic (proton) number on top & relative atomic mass in the bottom.

*Note: Mr & mass number isn't the same.

In the periodic table:

Group VIII noble gases have full outershell - stable.

the number of outer shell electrons is equal to the group number of that element

the number of occupied electron shells is equal to the period.

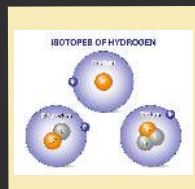
Isotopes

Isotopes are different atoms of the same element that have the same number of protons but different numbers of neutrons.

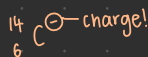
Isotopes of the same element have same chemical properties because they have the same number of electrons, therefore the same electronic configuration.

the isotopes symbol:

It can be also written with chemical symbol and dash the mass number. Eg: C-14



Carbon - 6 electrons needs 4 electrons to stabilize, so when this Carbon atom gains 4 electrons the overall number of electrons increase over protons so the charge is negative. eg:



relative atomic mass (RAM) is average mass of all isotopes & mass number isn't the same. RAM takes isotopes into account while calculating.

Calculating Relative atomic mass

Symbol: Ar

the average mass of the isotopes of an element compared to $1/12^{\text{th}}$ of the mass of an atom of ^{12}C .

relative atomic mass can be calculated from the mass number and relative abundances of all the isotopes using the equations:

$$A_r = \frac{(1 \text{ of isotope 1} \times \text{mass number of isotope 1}) + (2 \text{ of isotope 2} \times \text{mass number of isotope 2})}{100}$$

top line can be extended to include lots of isotopes of an element

metallic bonding

Metal atoms are held strongly by metallic bonding in giant metallic lattice.

within the metal lattice, the atoms lose electrons & become positively charged & the outer electrons no longer belong to a particular metal & are delocalized.

they move freely between the positive metal ions like a sea of electrons. metallic bonds are strong & are a result of attraction between positive metal ions & negative charged delocalized electrons.

Properties of metals

- high melting & boiling point - due to strong metallic bonds being hard to break
- conduct electricity - have free electrons to move & carry charge
- malleable & ductile - layers of positive ions can slide over each other & take different positions, strong but flexible so they can be made into specific shapes.

Chapter 2: Atoms, Elements and Compounds

Ions & Ionic bonds

- Ion is an electrically charged atom or group of atoms formed by loss or gain of electrons. - this is done by atom to stabilize.
- the loss/gain of electrons is done to gain full outer shell of electrons which is a more stable arrangement of electrons

Sodium atom sodium ion



Metal: all metals can lose electrons and become positively charged ions known as cations

Non-metal: non-metals can gain electrons from other atoms to become negatively charged ions known as anions.

there is space for 7 electrons so it is easier to lose the existing electron instead of gaining 7.

there is more protons than electrons now that sodium lost an electron. therefore it is positively charged.

Dot & Cross diagrams

- diagrams that show the arrangement of the outer-shell electrons in ionic or covalent compound or element.

Eg: Na & Cl - non-metal

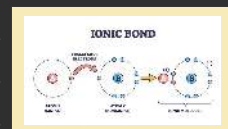


Lattice structure of ionic compounds

- ionic compounds have a giant lattice structure.
- lattice structures are arranged in an ordered & repeating fashion
- the lattices of ionic compounds have regular arrangement of alternating positive & negative ions.

Properties of ionic compounds

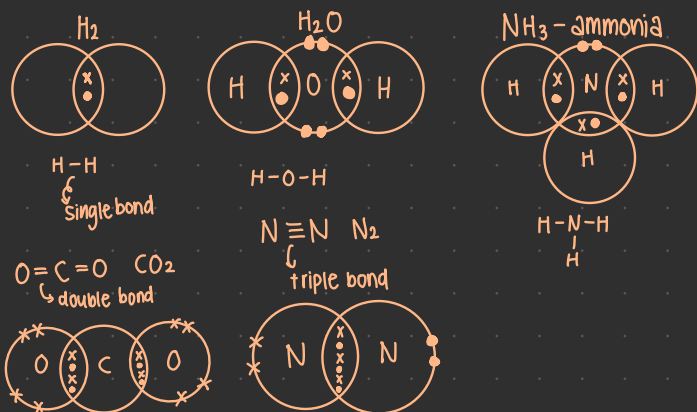
- usually solid at rtp
- high melting & boiling points because of strong electrostatic forces, a lot of energy will be needed to break bonds.
- good conductors of electricity in molten or in solution. this is because the particles are charged & freely moving
- poor conductors in solid state because they are in fixed positions within the lattice.



simple molecules and covalent bonds

- covalent compounds are formed when pairs of electrons are shared between atoms.
- Only non-metal elements participate in covalent bonding.
- when two or more atoms bond covalently, they are described as molecules.

* Dot & cross diagrams can be used to show covalent bonds eg:



Properties of simple molecular compounds

- small compound molecules are covalently bonded. they have low melting & boiling points so they are usually liquids or gases at room temperature.
- As molecule size increases, so does melting & boiling point.
- small molecules are poor electricity conductors.
- there is weak intermolecular forces so they have low melting & boiling point.
- intermolecular forces are weak so most small molecules are liquid/gas
- as molecule increases in size the intermolecular forces also increase this causes melting & boiling point to increase.
- * Most molecular compounds are poor conductors as there's no free ions to carry charge so most covalent compounds are insulators in solid state.

giant covalent structures



Diamond & Graphite

- allotropes of carbon with giant covalent structures, due to difference in bond arrangement they are physically different
- in diamonds, carbon atoms bond with 4 other carbons forming a tetrahedron.

all bonds are identical, strong & have no intermolecular forces.

in graphite, carbon atoms bond to 3 other forming layers of hexagons, leaving one free electron per carbon to become delocalized.

the covalent bonds within layers are very strong but the layers attract to each other with weak intermolecular forces.

PROPERTIES OF:

- **Graphite**
 - has free delocalized electrons so it can conduct electricity
 - layers are connected by weak forces so they can slide over each other so graphite is smooth & slippery
 - high melting & boiling point
 - less dense than diamond
 - used in pencils & lubricants
 - used to make inert electrodes
- **diamond**
 - insulator - no free electrons
 - high mp & bp - strong bonds
 - hard & dense
 - used in jewellery & cutting tools.

Silicon (IV) oxide

↳ macromolecular compound which occurs naturally as sand & quartz

- each oxygen atom covalent bonds with 2 silicon atoms & each silicon atom bonds with 4 oxygen atoms.
- tetrahedron is formed.

comparing diamond & silicon

- SiO₂ is hard & has a high melting point, insoluble in water & doesn't conduct electricity.
- SiO₂ is cheap as it's naturally available & used in making sandpaper and to line the insides of furnaces.

Chapter 3: Stoichiometry



Formulae

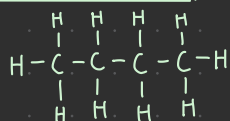
- each element has its own unique symbol as in the periodic table.
- Symbol contains two letters, first always capital & if there's a second, it will be in lowercase.

Eg: Sodium (Na)

- chemicals combine in fixed ratios that give full outer shells of electrons, chemical formulae tells the ratio of atoms eg: H_2O = 2 hydrogen atoms combined with 1 oxygen.
- structural formula tells the way in which the atoms in a particular molecule are bonded.
↳ done by a diagram or written version.
- molecular formulae tells you the actual number of atoms of each element in one molecule of the compound or element.

Example: Butane - a compound

Structural Formula



↑ diagram ↓ written: $CH_3CH_2CH_2CH_3$

Molecular Formula



Empirical Formula (ratio)



simplest whole number ratio of atoms of each element in a molecule
- organic molecules have different empirical & molecular formulae
↳ formula of ionic compound is always an empirical formula

Valency - the number of electrons to lose/gain for full outer shell
You can use valency to find chemical formula:

- aluminium sulfide

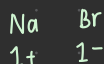
Al valency = 3 S valency = 2



Deducing Formulae of ionic compounds

- the valency can tell you charge, based on loss/gain eg: H needs to lose so it is 1^+ charge and Bromide needs to gain an electron so the charge is 1^- .
- the overall sum of charges must be 0.

Example: Sodium Bromide



- 1 sodium needed for one bromide
- NaBr.

Writing Word Equation & Symbols Equations.

Word equations

- show reactants & products using full chemical names
- arrow implies conversion of reactants to products.
- reaction conditions or catalysts can be written above the arrow.

Eg:

Sodium hydroxide + hydrochloric acid → Sodium chloride + water.

Names of compounds

- for compounds with metal & non-metal, the metal comes first & second atom has -ide at the end.
- if both are non-metals, and one is hydrogen, hydrogen comes first.
- for other non-metals, the element with lower group number comes first.
- for compounds with atoms that occur regularly eg: carbonate (CO_3^{2-}), hydroxide (OH^-), sulfate (SO_4^{2-}) and nitrate (NO_3^-). when these ions form compound with a metal atom, the metal name comes first.
- eg: KOH (Potassium hydroxide) & $CaCO_3$ calcium carbonate

Writing and balancing chemical equations.

- same number of atoms on each side needed!
- H, N, F, O, Cl, Br and I must be diatomic eg: H_2 & O_2 .
- groups of atoms like Nitrate (NO_3^-) that hasn't changed from one side to another, count it as a group not individual atoms.

State Symbols

- written after each formula to show Physical state of each substance.
- aqueous means substance is dissolved in water.
- eg: Solid (s)
liquid (l)
gas (g)
aqueous (aq)

Deducing Symbol equations.

Step #1: work out formula, and state symbols of reactants and products to construct an unbalanced symbol equation.

Step #2: balance the equation.

Balancing ionic equation.

- in aqueous solutions ionic compounds dissociate into their ions.

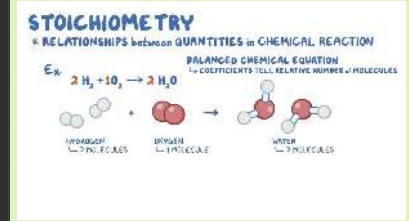
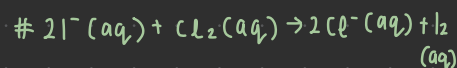
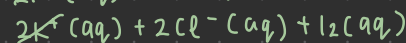
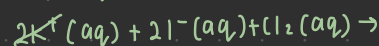
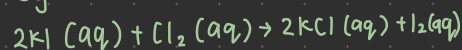
↳ separate into the component ions that formed them.

Step #1: Write balanced equation

Step #2: identify ionic substances & write down the ions separately

step #3: rewrite by eliminating ions that appear on both sides of the equation

Eg:



Chapter 3: Stoichiometry

Relative masses of atoms & molecules

- Relative atomic mass, A_r , is the average mass of isotopes of an element compared to $1/12^{\text{th}}$ of the mass of a ^{12}C atom.
- Relative molecular mass, M_r , is the sum of relative atomic masses, they are used for ionic compounds

Calculating Reacting Masses.

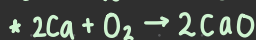
mass can't be created or destroyed.

Example:



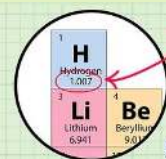
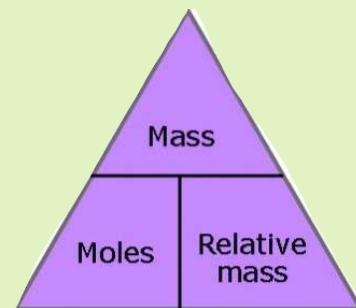
relative atomic mass of $\text{Ca} = 40$, $\text{O} = 16$

this means 80 units (2×40) of Ca react with (2×16) 32 units of oxygen



$$80 + 32 = 112$$

the ratio of mass will always be same regardless of unit eg: 80g Ca reacts with 32g of O will produce 112g of CaO . or 80 tonnes of Ca with 32 tonnes oxygen will produce 112 tonnes of Calcium oxide.



Relative Atomic Mass

Hydrogen $\rightarrow 1.007$

Carbon $\rightarrow 12.0107$

Oxygen $\rightarrow 15.9994$

Chlorine $\rightarrow 35.453$

[wiki How](#)

The mole and Avagadro's Constant

the mole, mol, is the unit of amount of substance that contains 6.02×10^{23} particles.

the number 6.02×10^{23} is known as Avagadro's Constant

$$\text{Amount of Substance (mol)} = \frac{\text{mass (g)}}{\text{molar mass (g/mol)}}$$

the mass of 1 mole of substance is known as molar mass - it is the same as A_r / M_r

$$\text{Mols} = \frac{\text{Grams (g)}}{\text{RAMs (relative atomic mass)}}$$

Reacting masses & limiting Reactants

chemical equations can be used to calculate the moles / mass of reactants & products. information in question is used to find the amount in moles then the ratio is identified through a balanced chemical equation.

Once moles are determined, they can be converted into grams using A_r or M_r .

* the reaction stops when a reactant is used up. the reactant used up first is the limiting reactant, it limits the duration & the amount of product produced. the product is directly proportional to the limiting reactant. the limiting reactant is the one not present in excess.

To perform reacting mass calculations:

- write balanced equation
- calculate moles of each reactant
- compare the moles & deduce the limiting reactant.

Concentration simply refers to amount of solute there is in a specific volume of solvent. The unit for concentration is g/dm^3 or mol/dm^3

$$\text{Concentration (g/dm}^3\text{)} = \frac{\text{mass of solute (g)}}{\text{Volume of solution (dm}^3\text{)}} \div M_r \text{ in grams}$$

$$1 \text{ dm}^3 = 1 \text{ litre}$$

$$\frac{\text{cm}^3}{1000} = \text{dm}^3 \quad \text{g/dm}^3 \quad \text{mol/dm}^3$$

$$\frac{\text{cm}^3}{1000} \div 1000 = \text{dm}^3 \quad \times M_r \text{ in grams}$$

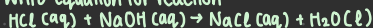
A common constant to use is: the molar gas volume is 24 dm^3 at room temperature & Pressure (R.T.P.) for gas calculations.

TITRATION CALCULATIONS

titration is a method of analysing the concentration of solutions. Acid-base titrations are one of the most important titration kind. You can be asked to calculate moles present in given amount, the concentration or volume required to neutralise acid/base.

Example:

Step #1: Write equation for reaction



Step #2: Calculate the moles for NaOH

Substance neutralizing the hydrochloric acid.

Step #3: Deduce the moles of acid

- use ratio, concentration & moles of NaOH

Step #4: Find concentration of acid

- use the mols & volume (given in Q).

EMPIRICAL FORMULA

- the simplest whole number ratio of the atoms of each element present in one molecule or formula unit
- organic molecules often have diff empirical & molecular formulae, formula of ionic compound is always empirical formulae

MOLECULAR FORMULA

- the actual number of atoms of each element present. to calculate molecular formula:

Step 1: find relative formula mass of empirical formula

Step 2: relative formula mass of molecular formula

relative formula mass of empirical formula

Step 3: multiply the number of each element present in the empirical formula by the number in step 2.

PERCENTAGE YIELD

Yield describes amount of product you get from reaction. You can never get 100% yield due to several reasons eg: reactant left behind in equipment, product lost during transfer, etc. Actual Yield is the recorded amount of product obtained. theoretical yield is the amount of product that would be obtained under perfect conditions.

$$\% \text{ Yield} = \frac{\text{actual Yield}}{\text{theoretical Yield}} \times 100$$

PERCENTAGE COMPOSITION BY MASS

$$\% \text{ of element} = \frac{\text{total mass of element in compound}}{\text{relative formula mass of compound}} \times 100$$

PERCENTAGE PURITY

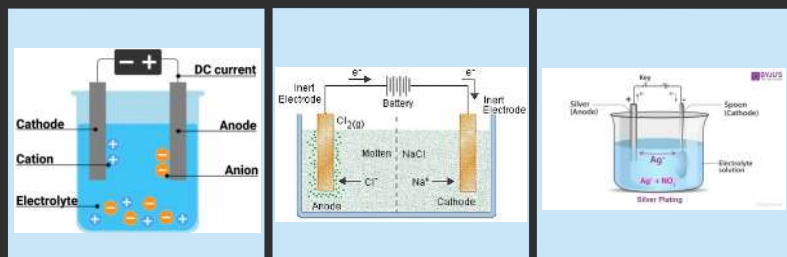
$$\% \text{ Purity} = \frac{\text{mass of pure substance}}{\text{total mass of substance}} \times 100$$

Chapter 4: Electrochemistry

Electrolysis



Electrolysis is the decomposition of an ionic compound, when molten or in aqueous solution, by the passage of electric current.



The transfer of charge during electrolysis includes:

- movement of electrons in external circuit
- loss or gain of electrons at the electrodes
- the movement of ions in the electrolyte.

During electrolysis, metals or hydrogen are formed at the cathode and that non-metals are formed at the anode.

Metal objects are electroplated to improve their appearance and resistance to corrosion.

Metals are electroplated by:

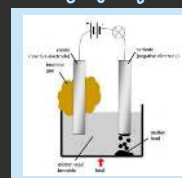
- #1: Place an object at the Cathode - the object to be electroplated.
 - #2: Place Pure metal at the anode - Metal will coat the object at Cathode.
 - #3: the electrolyte is a aqueous solution of soluble salt of pure metal at anode.
- * at anode metal loses electrons & form ion in solution, at cathode, tin gain electron & become tin atoms that deposit on the object, thus electroplating it.
- Electro plating is used to make metals resistant to corrosion & to improve the appearance

Identifying the products

BINARY COMPOUND IN MOLTEN STATE

Binary compound - compound with 2 different elements joined ionically. To predict their products made at electrodes, identify the ions. Positive ion to cathode, negative to anode, Cathode always metal product & anode has non-metal.

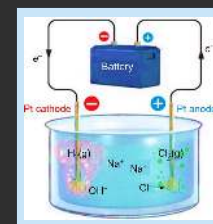
Eg: Lead(II) bromide
 Pb^{2+} Br^{-}
 anode - brown gas - bromine
 Cathode - grey metal - lead
 Potassium chloride
 K^{+} Cl^{-}
 anode - green/yellow gas - chlorine
 Cathode - metal build up - potassium



HALIDE COMPOUND IN CONCENTRATED/AQUEOUS SOLUTION

AQUEOUS NaCl
 - Brine is concentrated solution of aqueous sodium chloride
 - when electrolysed, gas is formed at both electrodes (chlorine & hydrogen)
 - both are important: chlorine for bleach, hydrogen for margarine (butter-like spread), sodium hydroxide for soap & detergents.

At Cathode
 H^{+} discharged (H^{+} gain electron)
 Hydrogen more reactive than sodium
 At Anode
 Cl^{-} gas discharged
 Cl^{-} lose electrons, Na^{+} & OH^{-} remain in solution & form NaOH
 Dilute H_2SO_4
 at cathode
 H^{+} at Cathode, Hydrogen gas released
 at anode
 OH^{-} at anode, Oxygen gas & water released



To identify gas released:
 oxygen - glowing splint test
 chlorine - Bleaches damp litmus paper to white
 Hydrogen - Pop squeak test

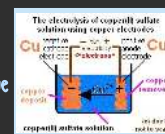
AQUEOUS COPPER(II) SULFATE WITH INERT & COPPER ELECTRODES

INERT (graphite / carbon)

oxygen at anode
 Carbon electrodes so the oxygen becomes CO_2 .
 Cu^{2+} at cathode

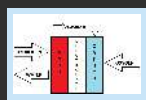
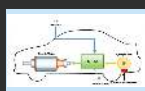
COPPER

Copper deposit on cathode
 Anode dissolves, copper is oxidized to ions & dissolve into the solution.



Hydrogen-oxygen Fuel Cells

A hydrogen-oxygen fuel cell uses hydrogen and oxygen to produce electricity with water as the only chemical product.



Advantages & Disadvantages of using hydrogen-oxygen fuel cells rather than gasoline / petrol vehicles:

Advantages

- * Don't Produce Pollution
- * release more energy
- * no power lost in transmission
- * Quieter, less noise

Disadvantages

- * fuel cells are expensive
- * difficult & expensive to store
- * fuel cells affected by low temperatures, less efficient.
- * less hydrogen filling stations

ionic half - equations

Oxidation: Substance loses electrons

Reduction: Substance gains electrons

- as the ions in the electrolyte come in contact with the electrodes, electrons aren't lost or gained and they form neutral substances and these are discharged as products at the end.

At the anode: -tive ions lose electrons & are oxidised

At the cathode: +tive ions gain electrons & are reduced

ionic half equations show oxidation & reduction of ions & result in balanced charges.

Eg: Molten Lead Bromine

at anode:
 $2Br^{-} \rightarrow Br_2 + 2e^{-}$

at cathode
 $Pb^{2+} + 2e^{-} \rightarrow Pb$

OIL
 Oxidation is loss of electrons
 RIG
 Reduction is gain of electrons

Dilute sulfuric acid

at anode
 $4OH^{-} \rightarrow O_2 + 2H_2O + 4e^{-}$

at cathode
 $2H^{+} + 2e^{-} \rightarrow H_2$

extra notes:

Chapter 5: Chemical energetics

exothermic reaction

exothermic reaction transfer thermal energy to the surrounding leading to an increase in temperature of the surroundings.

endothermic reaction

endothermic reaction takes in thermal energy from the surroundings leading to a decrease in the temperature of the surroundings

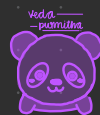
enthalpy change

- For atoms to react with each other in a chemical system they must come in contact through a collision
- Number of factors affect collisions: energy, orientation, number of collisions per second (frequency of collisions).
- there is a minimum amount of energy required for collision to be successful, that is for particles to react

transfer of energy

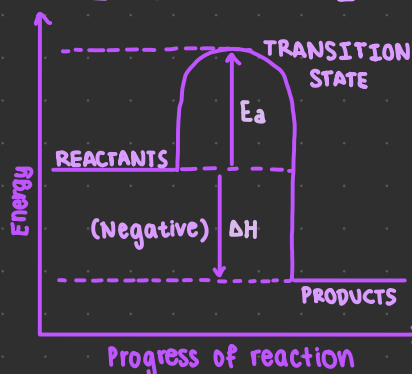
the transfer of thermal energy during a reaction is called the enthalpy change (ΔH) of the reaction, ΔH is negative for exothermic reactions and ΔH is positive for endothermic reactions

* Activation energy, E_a , is the minimum energy that colliding particles must have to react (successful collision)



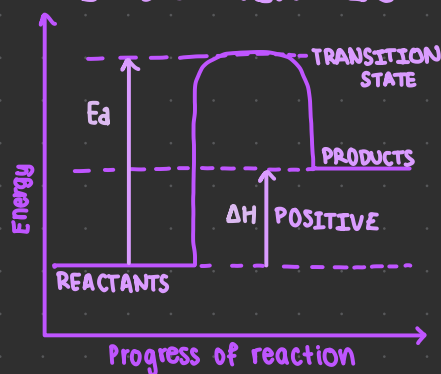
pathway diagrams

EXOTHERMIC



* Lower in energy than reactants

ENDOTHERMIC



* Higher in energy than reactants

bonds & enthalpy

bond breaking is an endothermic process and bond making is an exothermic reaction

the enthalpy change of a reaction

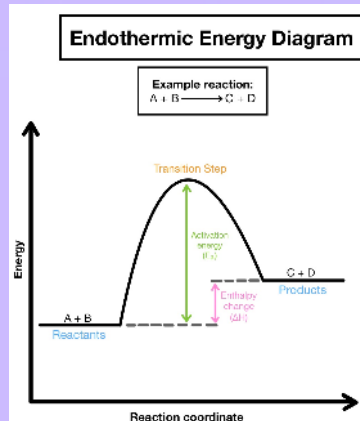
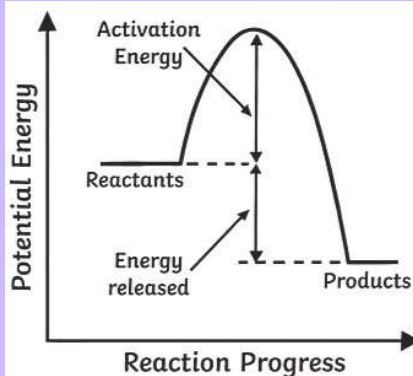
each chemical bond has specific bond energy associated with it, this is the amount of energy required to break the bond or amount of energy given out when the bond is formed. this energy can be used to calculate how much heat is released or absorbed during the reaction.

Method of calculating bonds

write a balanced equation if none are present.

add all the bond energies for the bonds in reactants - energy in
add all the bond energies for the bonds in the products - energy out.
calculate the enthalpy change

Enthalpy change (ΔH) = Energy in - Energy out.



Chapter 6: Chemical reactions



Physical & Chemical Changes

Physical Changes: easy to reverse, observable.
eg: melting, evaporating, etc.

Chemical changes: new chemical substance formed
very difficult to reverse

★ Exothermic - give out energy/heat

★ Endothermic - take in energy/heat



Rate of Reaction

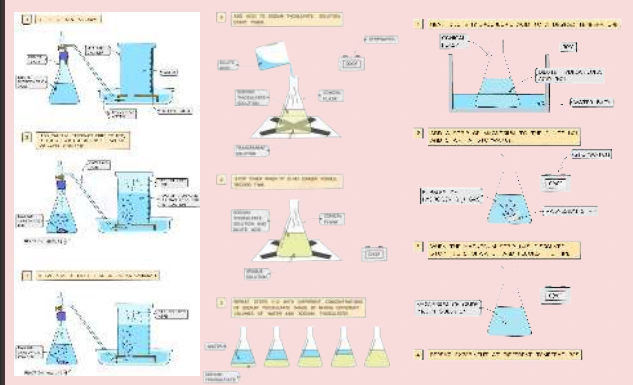
Collision theory

in order for a reaction to happen, particles must collide with each other and have sufficient energy to cause a successful collision - reaction. the minimum energy required for a successful collision is called Activation Energy (E_a)
the number of successful collisions depends on the number of particles per unit volume, the frequency of collisions and kinetic energy of particles. the particles must have sufficient activation energy.

A catalyst increases the rate of reaction of a reaction without being used up or changed. A catalyst decreases the activation energy E_a of a reaction.

Effect on the rate of reaction of:

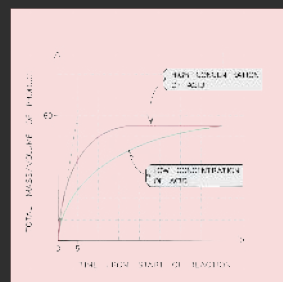
Concentration of solution - increased = higher rate of reaction
higher conc = more particles = more collision = faster reaction.
Pressure of gases - increased = higher rate of reaction
higher pressure = more collisions = faster reaction
Surface area of solid - increased = higher rate of reaction
increased = more place available for reaction to occur faster.
temperature - increased = higher rate of reaction
increased temp = more K.E = more collisions = fast reaction
Adding/removing catalyst - adding = speed up reaction
adding catalyst decreases E_a needed so more successful collisions lead to a faster reaction.



Experimenting

Method Choice	Advantage	Disadvantage
Discoloring cross measurement	Simple experiment that can be done with a variety of reactants	Difficult to see when the cross is completely gone, so the reaction may be overestimated
Collecting gas using a gas syringe	All the gas collected is from the reaction The syringe is easy to set up and keep going	Gas syringes are expensive and fragile If the syringe is not at the top of the flask, gas can be lost The gas can be lost if the syringe is not kept under the top of the flask quickly
Collecting gas using an inverted measuring cylinder	The experiment can be carried out in a beaker	The gas can be lost if the cylinder is not kept under the top of the flask quickly
Measuring the loss of mass of reactants	Simple experiment that can be done with a variety of reactants	Not suitable for gases with a small mass

Evaluating/Interpreting the Graphs



The steeper the curve, the faster the rate of the reaction

The curve is steepest initially so the rate is quickest at the beginning of the reaction

As the reaction progresses, the concentration of the reactants decreases and the rate decreases shown by the curve becoming less steep

When one of the reactants is used up, the reaction stops, the rate becomes zero and the curve levels off to a horizontal line

The amount of product formed in a reaction is determined by the limiting reactant

If the amount of limiting reactant increases, the amount of product formed increases

If the amount of the reactant in excess increases, the amount of product remains the same

Drawing a tangent to the slope allows you to show the gradient at any point on the curve

The steeper the slope, the quicker the rate of reaction

Redox

Roman numbers are used to indicate oxidation number (the number of electrons gained/lost during forming a chemical bond) of an element in compound eg: Iron(II) oxide
Iron loses 2 electrons

A redox reaction is a reaction that involves simultaneous oxidation and reduction
Oxidation is gain of oxygen & reduction is loss of oxygen.

↳ loss of electrons

↳ gain of electrons

increase in oxidation number

decreased oxidation number

Example: Zinc oxide + Carbon → Zinc + Carbon Monoxide
 $ZnO + C \rightarrow Zn + CO$

Zinc oxide has been reduced, it lost oxygen

Carbon has been oxidised, it gained oxygen

Zinc + Copper Sulphate → Zinc Sulphate + Copper

$Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$

$Zn(s) + Cu^{2+}(aq) + SO_4^{2-}(aq) \rightarrow Zn^{2+}(aq) + SO_4^{2-}(aq) + Cu(s)$

Spectator ion: remove SO_4^{2-} because it doesn't change.

$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$

★ Write half equations:

$Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$ — Oxidised, lost electrons

$Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$ — Reduced, gained electrons

- the oxidation number of elements in their uncombined state is zero
- the oxidation number of monatomic ion is the same as the charge on ion
- the sum of oxidation numbers in a compound is zero.
- the sum of oxidation number in a ion is equal to the charge of the ion.

Identifying Redox reaction by colour changes.

Acidified potassium manganate
 $KMnO_4$ - Oxidising agent

used to test for presence of reducing agent.

when acidified potassium manganate is

added to a reducing agent, the colour changes from Purple to colourless.

Potassium iodide

KI - reducing agent

used to test for oxidising agents

when potassium iodide is added to an oxidising agent it colour changes from colourless to red-brown colour.

Oxidising Agent: a substance that oxidises another & reduces itself.

Reducing Agent: a substance that reduces another & oxidises itself.

Example:

Copper (II) oxide + Hydrogen → Copper + Water

$CuO + H_2 \rightarrow Cu + H_2O$

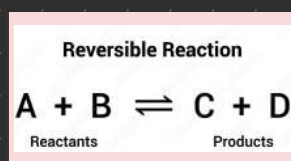
$Cu^{2+} + 2e^- \rightarrow Cu$ ~ gains electrons, reduced

loses electrons
oxidised
Reducing agent: $H_2 \rightarrow 2H^+ + 2e^-$

oxidising agent.

Chapter 6: Chemical reactions

Reversible & Equilibrium



Some chemical reactions are reversible and are shown with the symbol $\rightleftharpoons$.

A reversible reaction in a closed system is at equilibrium when the rate of the forward reaction is equal to the rate of the reverse reaction, the concentration of reaction and products are no longer changing.

the position of equilibrium is affected by changing temperature because increasing temp will shift equilibrium in the endothermic direction, increased Pressure shifts equilibrium in the direction that produces less molecules of gas, increased Concentration shifts equilibrium to the right to reduce the effect of increase, a catalyst does not affect the position of equilibrium but it increases the rate at which equilibrium is reached.

Copper (II) Sulfate

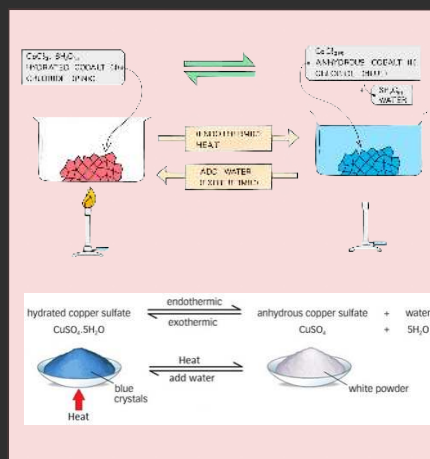
the effect of heat on hydrated $CuSO_4$
it loses water of crystallization as result of evaporation and the crystals turn blue to white and from crystalline form to amorphous (no clear shape/form).

Addition of water to anhydrous $CuSO_4$

Anhydrous copper sulphate is a white powder that turns blue in contact with water

$$CuSO_4 \cdot 5H_2O \xrightleftharpoons[\text{exothermic}]{\text{endothermic}} CuSO_4 + 5H_2O$$

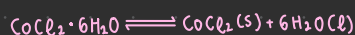
hydrated copper sulfate anhydrous copper sulfate



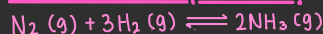
Cobalt (II) Chloride

the effect of heat on hydrated $CoCl_2$
when heated, the pink crystals turn to blue anhydrous Cobalt (II) Chloride.

Addition of water to anhydrous $CoCl_2$
anhydrous $CoCl_2$ is blue and when water is added, it turns pink.



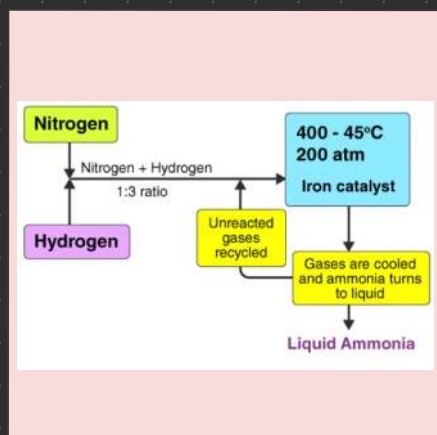
The Haber process



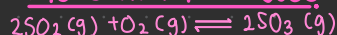
- * The haber process is used to produce ammonia for industrial use, Ammonia is used for fertilizers, manufacturing plastic and etc.
- * the Nitrogen in haber process is from the air and hydrogen from methane gas
- * the typical conditions for the haber process is
 - $450^\circ C$ - forward is exothermic, increasing temp shifts position to left & lowers yield.
 - $20\,000\,kPa / 200\,atm$ to shift equilibrium to right & too high pressure is dangerous
 - iron beads are used as a catalyst. helps reach equilibrium faster.

The Stages of haber process:

- Step #1: H_2 & N_2 obtained through natural gas and pumped into compressor through pipe
- Step #2: the gases are compressed to $200\,atm$ inside the compressor.
- Step #3: the pressurized gases are pumped into tank with iron beads & temp of $450^\circ C$
- Step #4: Some H_2 & N_2 react, the unreacted gas & ammonia pass to cooling tank & ammonia is liquidified and removed to storage vessels.
- Step #5: the unreacted H_2 & N_2 is recycled back into system to restart.



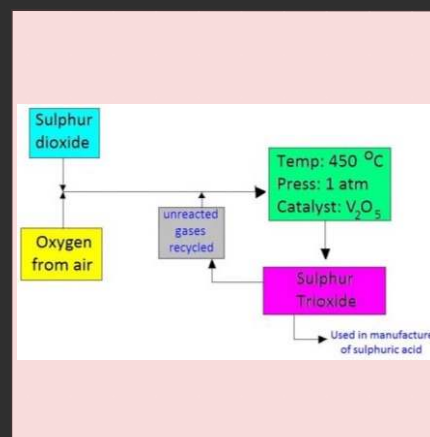
The Contact Process



- * the contact Process is the conversion of sulfur dioxide to sulfur trioxide because it is used for fertilizers, soaps and detergents.
- * the first Stage is the production of sulfur dioxide, either by burning sulfur to oxidise or roasting sulfide ores

$$S + O_2 \rightarrow SO_2$$
- * the main stage is the oxidation of sulfur dioxide to sulfur trioxide using a vanadium (V) oxide, V_2O_5 , catalyst

$$2SO_2 + O_2 \rightleftharpoons 2SO_3$$
- * the oxygen is obtained from air
- * the typical conditions for contact Process is:
 - $450^\circ C$, higher temp cause lower yield of sulfur trioxide
 - $200\,kPa / 20\,atm$, any higher pressure causes SO_2 to liquefy
 - vanadium (V) oxide



Chapter 7: Acids, bases and salts

the characteristics of acids & bases

the characteristic properties of acids in reactions with:

Metal – only metals above hydrogen in reactivity series will react with dilute acids, they form a salt & hydrogen gas. the name of the salt depends on the anion in the acid.

Carbonates – corresponding metal salt; CO_2 & water will be formed.

Bases – metal oxides & metal hydroxides are bases, when react, a neutralization reaction occurs. A salt & water will be produced.

Acids & base effect on:

litmus paper
thymolphthalein
methyl orange

indicator	acidic solution	alkaline solution
litmus	red	blue
thymolphthalein	colorless	blue
methyl orange	red	yellow

Bases are oxides or hydroxides of metals and **alkalis** are soluble bases.

Base reaction with:

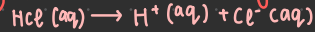
Ammonium salts – the ammonium salt decompose when warmed with a base, ammonia is volatile & easily displaced from salt by another alkali so a salt, water and ammonia is formed.

Aqueous solutions of acid contain H^+ ions and aqueous solutions of alkalis contain OH^- ions.

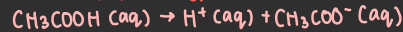
Acids are proton donors, Bases are proton acceptors.

A strong acid is an acid that is completely dissociated in aqueous solution and a weak acid is an acid that is partially dissociated in aqueous solution.

Hydrochloric acid is a strong acid:



Ethanoic acid is a weak acid:



Neutralisation Reactions

- When acids are added to water, they form positive charged hydrogen ions (H^+)
- When alkalis are added to water, they form negative charged hydroxide ions (OH^-)
- Neutralisation occurs when acid reacts with an alkali, the H^+ & OH^- ions react to produce water H_2O
- $\text{H}^+ (\text{aq}) + \text{OH}^- (\text{aq}) \rightarrow \text{H}_2\text{O (l)}$



Colour & pH with Universal Indicator paper

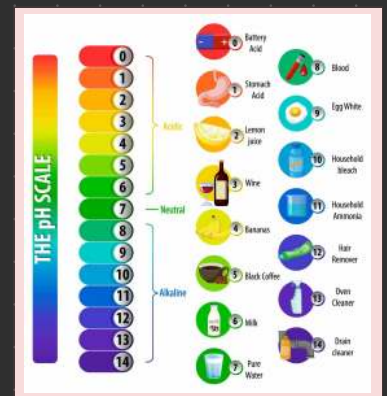
the **pH scale** is a numerical scale which is used to show how acidic or alkaline a solution is.

the scale goes from most acidic (1) to most alkaline (14). All pH below 7 is acids, above 7 is alkaline and 7 is neutral.

The more H^+ ions, the stronger the acid, lower pH. The more OH^- ions, the stronger the alkali, higher pH.

The pH scale is logarithmic meaning each change of 1 on the scale represents change by factor of 10.

The **Universal indicator** is a mixture of different indicators to measure pH.



Preparation of salts

the preparation, separation, and purification of **soluble salts** by reaction of an acid with:

an alkali by titration: $\text{acid} + \text{alkali} \rightarrow \text{salt} + \text{water}$

- Carry out titration to work out the precise volume of acid and alkali for neutralisation to occur.
- Mix the appropriate volumes of acid and alkali so we are left with solution containing salt & water.
- Evaporate the water by gently heating the solution allow to cool and the soluble salt will crystallise.
- Filter the solid salt, and dry.

Excess metal: $\text{acid} + \text{metal} \rightarrow \text{salt} + \text{hydrogen}$

any metal more reactive than hydrogen [Fe , K , Na , Ca , Mg , Al , Zn , etc] reacts with acid to form a salt & hydrogen gas.

Pour acid into beaker, add metal strip & keep adding until metal is in excess, remove excess by filtration, evaporate the filtrate till salt is left behind, & dry.

Insoluble bases: $\text{acid} + \text{base} \rightarrow \text{salt} + \text{water}$

add excess metal oxide to acid in beaker, warm the beaker and filter out excess metal oxide, evaporate the water and dry the salt.

Insoluble carbonate: $\text{acid} + \text{carbonate} \rightarrow \text{salt} + \text{water} + \text{carbon dioxide}$

add acid to beaker, warm it, add carbonate until it is in excess, filter the excess, evaporate the filtrate until crystallisation point, dry the salt.

Preparation of insoluble salt with the precipitation method

Place a known volume of soluble salt to a beaker keep adding another soluble salt and stir to mix until no more precipitate forms. filter the precipitate, the residue is the insoluble salt. wash residue with water and dry the salt.

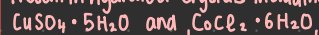
Solubility rules for salts:

- sodium, potassium & ammonium salts are soluble
- nitrates are soluble
- chlorides are soluble, except lead & silver.
- Sulfates are soluble, except barium, calcium & lead
- carbonates are insoluble except sodium, potassium & ammonium
- hydroxides are insoluble except sodium, potassium, ammonium and calcium (partially).

A **Hydrated Substance** is a substance that is chemically combined with water

An **Anhydrous Substance** is a substance that is containing no water.

the term **water of crystallisation** is the water molecules present in hydrated crystals including



↳ indicates 6 water molecules are attached / bound to the cobalt chloride.

	Soluble Salts	Insoluble Salts
Carbonates	Sodium carbonate Zinc carbonate Ammonium carbonate	All other carbonates
Nitrates	All nitrates	None
Chlorides	All chlorides except	Lead(II) chloride Silver chloride
Sulfates	All sulfates except	Barium sulfate (slightly soluble) Lead(II) sulfate Bismuth sulfate
Ammonium salts & Group I metal salts	All	None

Oxides

Oxides are compounds made from one or more atoms of oxygen combined with one other element. Eg: SO_2 , CO_2 , CuO , MgO .

Oxides can be classified as acidic or base based on whether they are bonded to a metal or nonmetal. Some are neutral oxides like N_2O , NO & CO .

Acidic Oxide	Formula	Acid Produced with Water
sulphur trioxide	SO_3	sulphuric acid, H_2SO_4
sulphur dioxide	SO_2	sulphurous acid, H_2SO_3
carbon dioxide	CO_2	carbonic acid, H_2CO_3
phosphorus(V) oxide	P_2O_5	phosphoric acid, H_3PO_4

Basic Oxide	Formula
magnesium oxide	MgO
sodium oxide	Na_2O
calcium oxide	CaO
copper(II) oxide	CuO

Acidic Oxides

- * formed when a non-metal combines with oxygen.
- * when dissolved in water they produce acidic solution eg: CO_2 , SO_2 , NO_2 .

Basic Oxides

- * formed when a metal combines with oxygen
- * eg: CuO & CaO .

Amphoteric Oxides

- * react with acids & bases to produce salt & water
- * eg: Al_2O_3 and ZnO

Chapter 8: The periodic table

arrangement of elements

the periodic table is an arrangement of elements & groups in order of increasing proton number / atomic number.
vertical columns are called groups: I-VII & Group 0 (not VIII)
horizontal rows are called periods: 1-7

Metallic characteristics decrease across a period, left to right and increase down a period
Metals occur on the left hand side of the periodic table and non-metals on right.
Between them lie the metalloids / semi-metals.

Electronic configuration will show the number of occupied shells which is the same as the period & the number of electrons on the last shell is the group number.

group vii properties

Halogens, the diatomic non-metals with general trends down the group such as:

- increasing density
- decreasing reactivity

The appearance of halogens at r.t.p:

- Chlorine (Cl): Pale yellow-green gas
- Bromine (Br): red-brown liquid
- Iodine (I): grey-black solid

In displacement reactions, the more reactive halogen atoms oxidise the less reactive halide ions, causing halide ions to lose electrons and form halogen atoms.

the halogen atoms then gain electrons to form halide ions, which are reduced. Redox reaction.

transition elements

transition elements are metals that:

- have high density
- high melting point
- form coloured compounds
- often act as catalysts as elements and in compounds.

transition elements have ions with variable oxidation numbers, including iron (II) & iron (III)

noble gases

* the noble gases are the unreactive, monatomic gases

* they all have a full outer shell of 8 electrons
So they are very stable and unreactive.

group i properties

Alkali metals, relatively soft metals with trend down the groups:

- decreasing melting point
- increasing density
- increasing reactivity



Chapter 9: Metals

alloys and their Properties

Properties of metals

- * conduct heat & electricity
- * Malleable & ductile
- * high density & high melting point

Properties of non-metals:

- * don't conduct heat & electricity
- * brittle & easily break
- * low density & low melting point.

Metals reaction to:

- dilute acids - Salt + hydrogen formed.
- Cold water & steam - metal hydroxide + hydrogen
- Oxygen - some metal don't react eg: gold
Others form a metal oxide.

An alloy is a mixture of metal with other elements including:

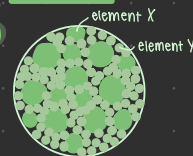
brass - copper + zinc

Stainless Steel - iron, and other elements like chromium, nickel and carbon.

Alloys can be harder, stronger, than pure metals and more useful. This is because the different sized atoms from different elements means the layers can't slide over each other.

Brass - Made of Cu & Zn but stronger than both, used for musical instruments, ornaments & door knobs.

Stainless Steel - used in cutlery as it is hard & is resistant to rusting.



Corrosion of metals

the conditions required for rusting of iron and steel to form hydrated iron(III) oxide are iron, water and oxygen.

Some common barrier method to prevent rusting, including Painting, greasing and coating with plastic.

Sacrificial Protection is the process of using a more reactive metal can be attached to a less reactive metal so the more reactive metal will corrode first. Zinc is used to galvanise.

All the barrier methods prevent the iron from being exposed to air and oxygen.

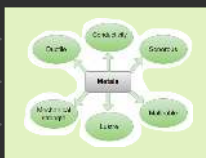
Galvanising is the process of using zinc to coat the iron. Zinc is more reactive so it will lose electrons more easily and oxidise / rust faster, so the zinc is sacrificed.

Uses of metals

Aluminium in the manufacture of aircraft cause of low density.
Aluminium in the manufacture of overhead electrical cables cause of low density and good electricity conductivity.

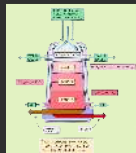
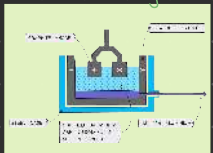
Aluminium in the food containers because of resistance to corrosion.

Copper in electrical wiring due to good conduction and ductility.



Extraction of metals

the ease of obtaining metals from ore depends on their position of the metals in the reactivity series.



The extraction of iron from hematite ore in the blast furnaces takes place by:

- the burning of carbon (Coke) to provide heat & produce Carbon dioxide
 $C + O_2 \rightarrow CO_2$
- the reduction of Carbon dioxide to carbon monoxide
 $C + CO_2 \rightarrow 2CO$
- the reduction of iron (III) oxide by carbon monoxide
 $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$
- thermal decomposition of Calcium Carbonate / limestone to produce Calcium oxide
 $CaCO_3 \rightarrow CaO + CO_2$
- the formation of slag
 $CaO + SiO_2 \rightarrow CaSiO_3$

The main ore of aluminium is bauxite and aluminium is extracted by electrolysis.

Bauxite is first purified to produce Al_2O_3 (aluminium oxide)

Aluminium oxide is then dissolved in molten cryolite

- this is because Al_2O_3 has a high melting point which needs a lot of energy but the mixture has lower melting point without interfering the reaction.
the mixture is placed in electrolysis cell and the carbon anodes are regularly replaced.

- this is because the carbon reacts with oxygen in Al_2O_3 & are burnt away over time the electrode is made from steel & lined with graphite.

At the cathode: $Al^{3+} + 3e^- \rightarrow Al$

At the anode: $2O^{2-} \rightarrow O_2 + 4e^-$

Reactivity series

The order of Reactivity is:

Potassium, Sodium, calcium, magnesium, aluminium, Carbon, Zinc, iron, hydrogen, copper, silver, gold.

Metal atoms form positive ions by loss of electrons when they react with other substances.

The tendency to lose electrons is a measure of how reactive the metal is, a metal high up on series loses electrons more easily than the ones below it.

Displacement reactions occur when a more reactive metal displaces a less reactive metal from a compound the more reactive metals lose electrons easier, making them better reducing agents.

Example:



Metals	Reactivity
Potassium	Reacts with water
Sodium	
Lithium	
Barium	
Strontium	
Calcium	Reacts with acids
Magnesium	
Aluminium	
Manganese	
Zinc	
Chromium	
Iron	
Cadmium	
Cobalt	
Nickel	
Tin	Included for comparison
Lead	
Hydrogen	Highly unreactive
Antimony	
Bismuth	
Copper	
Mercury	
Silver	
Gold	
Platinum	

Aluminium is high in the reactivity series but in reality it doesn't react with water & slow with dilute acids. This is because it reacts readily with oxygen to form a protective layer of aluminium oxide, this layer prevents reaction with water & acids.

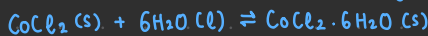
Chapter 10: Chemistry of the environment

Testing for the Presence of Water:

Anhydrous cobalt (II) chloride

cobalt (II) chloride turns blue to pink in contact with water

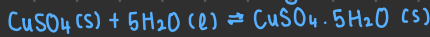
equation: anhydrous cobalt (II) chloride + water $\rightleftharpoons$ hydrated cobalt (II) chloride



Anhydrous copper (II) sulfate

copper (II) sulfate turns white to blue in contact with water, the equation is:

Anhydrous copper (II) sulfate + water $\rightleftharpoons$ hydrated copper (II) sulfate



Testing the Purity of water

Water has boiling point of 100°C & melting of 0°C

impure water will have higher boiling point & will melt below 0°C .

* Distilled water is used in practical rather than tap water because it has a few chemical impurities.

water

Water from natural sources include substances like:

- dissolved oxygen, metal compounds, plastics, sewage, harmful microbes, nitrates from fertilizers, phosphates from fertilizers & detergents.

Some of these substances are useful because:

- dissolved oxygen used by aquatic life for respiration
- Some metal compounds provide essential minerals for life.

Some of these substances are potentially harmful like:

- Some metal compounds are toxic
- Some plastics harm aquatic life
- Sewage contains harmful microbes that cause diseases
- nitrates & phosphates cause deoxygenation of water & harm life.

The treatment of domestic water supply:

step #1: Sedimentation & filtration to remove solids

step #2: use of carbon to remove tastes & flavours

Step #3: Chlorination to kill all the microbes



fertilisers

* ammonium salts & nitrates are used as fertilizers

- NPK fertilizers provide elements nitrogen, phosphorus & potassium for improved plant growth.

* Nitrogen - chlorophyll & protein for healthy leaves

* Phosphorus - promote healthy roots

* Potassium - promote growth of healthy fruit & flower.

Common fertilizer compounds:

ammonium nitrate (NH_4NO_3)

potassium sulfate ($(\text{NH}_4)_3\text{PO}_4$)

ammonium phosphate (K_2SO_4)



air quality and climate

The composition of clean dry air has approx. 78% nitrogen, N_2 , 21% oxygen, O_2 , & the remainder is a mixture of noble gases & carbon dioxide, CO_2 .

The sources of air pollutants:

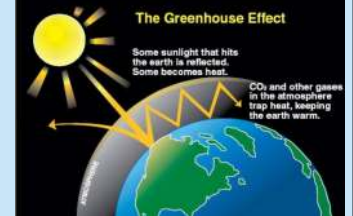
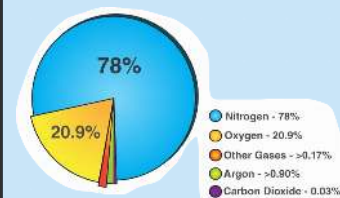
- CO_2 from complete combustion of carbon containing fuels
- CO & particulates from incomplete combustion of carbon containing fuels
- methane from decomposition of vegetation & waste gas from animal digesting
- oxides of nitrogen from car engines
- Sulfur dioxide from combustion of fossil fuels which contain sulfur compounds.

The air pollutants have adverse effects such as:

- carbon dioxide leads to increased global warming & lead to climate change
- Carbon monoxide is a toxic gas
- Particulates cause increased risk of respiratory problems & cancer
- methane leads to increased global warming & lead to climate change
- oxides of nitrogen cause acid rain, photochemical smog & respiratory issues.
- Sulfur dioxide causes acid rain.

The greenhouse effect:

the greenhouse gases like CO_2 , methane, nitrous oxide & etc absorb heat that is reflected off ground from sun, and this absorbed heat is re-emitted into all directions, this reduces thermal energy lost to space & increases temp on earth leading to global warming & climate change.



strategies that reduce the effect of these environmental issues are:

- For climate change: Planting trees, reduction in livestock farming, decreasing the use of fossil fuels & using renewable energy.
 - For Acid rain: Use catalytic converters in vehicles, reduce emissions of sulfur dioxide by using low-sulfur fuels & flue gas desulfurisation with calcium oxide.
 - Oxides of nitrogen form in car engines & are removed by catalytic converters by mixing with carbon monoxide. $2\text{CO} + 2\text{NO} \rightarrow 2\text{CO}_2 + \text{N}_2$
- Photosynthesis is the reaction between carbon dioxide & water to make glucose and oxygen in the presence of chlorophyll and using energy from light.

Word Equation: Carbon dioxide + Water $\rightarrow$ Glucose + Oxygen

Chemical Equation: $6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$



Veda —
— purmitha

naming organic compounds

eg:
$$\begin{array}{ccccccc} & \text{H} & & \text{H} & & \text{H} & & \text{H} \\ & | & & | & & | & & | \\ \text{H} & - \text{C} & - & \text{C} & - & \text{C} & - & \text{C} & - & \text{H} \\ & | & & | & & | & & | \\ & \text{H} & & \text{H} & & \text{H} & & \text{H} \\ & & & & & | & & \\ & & & & & \text{H} & - & \text{C} & - & \text{H} \\ & & & & & & & | \\ & & & & & & & \text{H} \end{array}$$

 C_5H_{12}

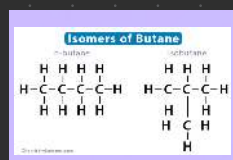
- have same functional group
- have same general formula
- differing from one member to next by a $-CH_2-$ unit
- displaying a trend in physical properties
- sharing similar chemical properties.

Unsaturated compound has molecules in which one or more carbon-carbon bonds are not single bonds.

eg:
 $\text{CH}_2 = \text{CH}_2$

$$\text{CH}_3\text{CH}_2\text{OH}, \text{CH}_3\text{COOCH}_3$$
$$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3 \text{ and } \text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_3$$

C_4H_8 as
 $CH_3CH_2CH=CH_2$ and
 $CH_3CH=CHCH_3$



FIRST PART OF NAME		SECOND PART OF NAME		Alcohol	Structural formula	Displayed formula
NAME	NUMBER OF CARBON ATOMS	NAME	FUNCTIONAL GROUP			
METH...	1	...AN-	NOF	ALCOHOL		
ETH...	2	...AN-	NOF	ALCOHOL		
PROP...	3	...AN-	NOF	ALCOHOL		
BUT...	4	...AN-	NOF	ALCOHOL		
PENT...	5	...AN-	NOF	ALCOHOL		
HEX...	6	...AN-	NOF	ALCOHOL		
ETHANE ETHENE						
ETHANOL ETHANOIC ACID						

NAME OF ALCOHOL	NAME OF CARBOXYLIC ACID	NAME OF ESTER
ETHANOL	METHANOIC ACID	ETHYL METHANOATE
BUTANOL	METHANOIC ACID	BUTYL METHANOATE
PENTANOL	ETHANOIC ACID	PENTYL ETHANOATE

$$R_1-\overset{\text{OH}}{\underset{\text{OH}}{\text{C}}}-\text{OH} + R_2-\text{OH} \longrightarrow R_1-\overset{\text{O}}{\underset{\text{O}}{\text{C}}}-\text{O}-R_2 + \text{H}_2\text{O}$$

Note: R may be a H atom, alkyl/aryl group!

For other alkenes, to find their single monomer, break their double bonds.

↓

$\left[\text{C}(=\text{O}) - \text{C}_6\text{H}_4 - \text{C}(=\text{O}) - \text{O} - \text{C}_6\text{H}_4 - \text{O} \right]$
a polyester (PET)

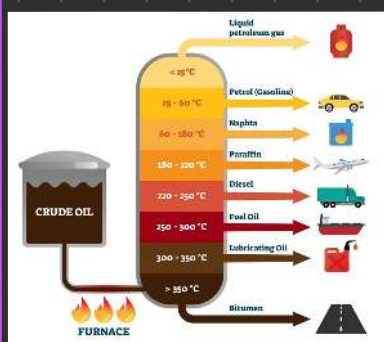
$$\left[\text{N} \begin{array}{c} \text{H} \\ | \end{array} \text{---} \text{C} \begin{array}{c} \text{O} \\ || \end{array} \text{---} \text{N} \begin{array}{c} \text{H} \\ | \end{array} \text{---} \text{C} \begin{array}{c} \text{O} \\ || \end{array} \right]_n$$

Chapter 11: Organic Chemistry

Fuels



Fossil fuels: coal, natural gas & petroleum
Methane is the main component in natural gases
Hydrocarbons are the compounds that contain hydrogen and carbon only. Eg: CH_4 (methane)
Petroleum is a mix of hydrocarbons, the petroleum is separated into useful components by fractional distillation
The properties of fractions obtained change from the bottom to top of fractionating column by:
decreasing chain length, higher volatility, lower boiling points, lower viscosity.



FRACTION	NUMBER OF CARBON ATOMS	BOILING POINT RANGE / °C	BOILING POINT & VISCOSITY INCREASE GOING DOWN
LIQUID PETROLEUM GAS	1-4	25-40	
GASOLINE / PETROL	4-12	40-100	
NAPHTHA	12-14	100-150	
KEROSENE / PARAFFIN	12-16	150-240	
DIESEL / GASOL	16-20	240-300	
FUEL OIL	18-25	250-320	
LUBRICATING OIL	25-40	300-350	
BITUMEN	MORE THAN 40	MORE THAN 350	

alcohols

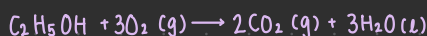
- alcohol is also called ethanol ($\text{CH}_3\text{CH}_2\text{OH}$).
it is formed by:
Fermentation of aqueous glucose (Plant sugar) at $25-35^\circ\text{C}$ in the presence of yeast and in the absence of oxygen. Ethanol can also be made by the Catalytic addition of Steam to ethene at 300°C and 6000 kPa / 60 atm in the Presence of an acid catalyst.

ADVANTAGES & DISADVANTAGES OF :

* FERMENTATION
- less energy
- Cheap
- renewable raw materials
- labour intensive
- time consumive

* CATALYTIC ADDITION OF STEAM TO ETHENE
- Continuous process
- less labour needed
- fast process
- expensive
- high energy needed
- non-renewable raw materials

COMBUSTION OF ETHANOL



they combust completely in the presence of water, When oxygen is Scarce, they produce water and carbon monoxide or Carbon (Soot).

Ethanol can be used as a :

- Solvent
- fuel

alkanes

ALKANES & ALKENES ARE TYPES OF HYDROCARBONS

the bonding in alkanes is single covalent and Alkanes are saturated hydrocarbons

- 1 carbon atom - methane (CH_4)
- 2 carbon atoms - ethane (C_2H_6)
- 3 carbon atoms - Propane (C_3H_8)
- 4 carbon atoms - Butane (C_4H_{10})
- 5 carbon atoms - Pentane (C_5H_{12})

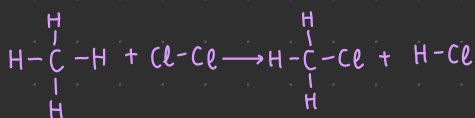
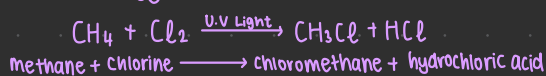
Alkanes are generally unreactive (with exception to Combustion & Chlorine substitution).

Substitution reaction is where one atom or group of atoms is replaced by another atom or a group of atoms.

Alkane	Structural formula	Displayed formula
Methane	CH_4	
Ethane	CH_3CH_3	
Propane	$\text{CH}_3\text{CH}_2\text{CH}_3$	
Butane	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$	

SUBSTITUTION OF ALKANES WITH CHLORINE

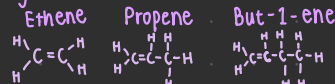
This is a photochemical reaction where ultraviolet light provides the activation energy (E_a).



alkenes

bonding in alkenes including a double carbon-carbon covalent bond and Alkenes are UNSATURATED Hydrocarbons

Eg:



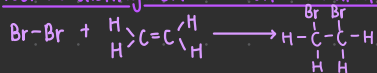
MANUFACTURING ALKENES

a process of catalytic cracking is used to convert long chain molecules into short-chain and more useful hydrocarbons.

Short chain alkenes are formed by cracking long chained alkenes.

Cracking is done by heating the hydrocarbon molecules to $600-700^\circ\text{C}$ to vapourise them. the vapours pass over hot powdered catalyst like aluminum / silica & this breaks their covalent bonds causing thermal decomposition. This is done cause short chained alkenes are more useful hydrocarbons

Test to distinguish between saturated / unsaturated hydrocarbons



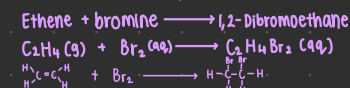
(Brown water)
Bromine water + Ethene (unsaturated)
 $\rightarrow$ 1,2-dibromoethane (colourless)

if colour doesn't change, its ALKANE.

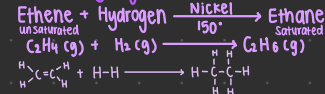
In an addition reaction, only one Product is formed.

PROPERTIES OF ALKENES:

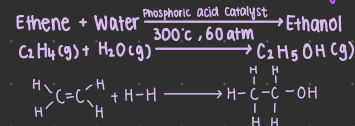
• Alkene + bromine / aqueous bromine



• Alkene + hydrogen in presence of nickel catalyst



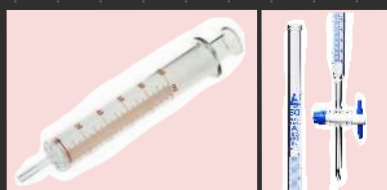
• Alkene + steam in presence of acid catalyst



Alkene	Structural Formula	Displayed Formula
Ethene	$\text{CH}_2=\text{CH}_2$	
Propene	$\text{CH}_3-\text{CH}=\text{CH}_2$	
But-1-ene	$\text{CH}_3-\text{CH}_2-\text{CH}=\text{CH}_2$	
But-2-ene	$\text{CH}_3-\text{CH}=\text{CH}-\text{CH}_3$	

Chapter 12: Experimental techniques & Chemical analysis

- **Stopwatches** – measure time
- **thermometers** – measure temperature
- **balances** – measure weight
- **burettes** – measure volume of liquid or a gas
- **volumetric pipettes** – to transfer specific amount of liquid
- **measuring cylinder** – measure liquid volume (less accurate than) burette
- **gas syringe** – insert or withdraw specific volume of gas.



experimental design

Advantages and disadvantages of methods & apparatus

Apparatus choice	Advantage	Disadvantage
Temperature probe versus thermometer	• Easy to use and can be used to take readings at specific locations • Can be used to monitor temperature over a long period of time	• Probe can be damaged by some reagents • High cost of probe and its use
Volumetric pipette versus measuring cylinder	• Pipette measures very 'exact' volume	• Harder to use and only measures one fixed volume
Gas syringe versus measuring cylinder for collecting gases	• Gas syringe allows easy collection and keeps the gas dry	• Limited volume range and scale • Difficult to use and expensive
Microscale versus normal scale quantities	• Less wasteful of reagents • Lower energy • Better	• Harder to see what's happening • Use a lot of material in reagents and purifying products

Solvent is a substance that dissolves a solute

Solute is a substance that is dissolved in a solvent

Solution is a mixture of one or more solutes dissolved in a solvent

Saturated solution is a solution containing the maximum concentration of a solute dissolved in the solvent at a specified temperature.

Residue is a substance that remains after any evaporation, distillation, filtration, or any similar process

Filtrate is a liquid / solution that has passed through a filter.



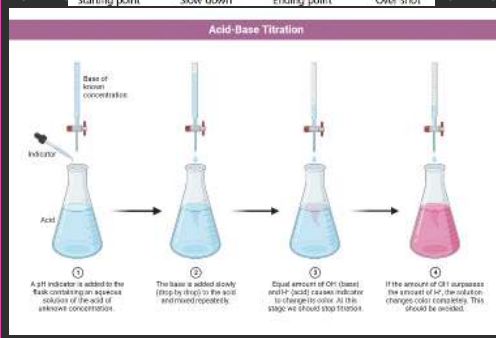
acid base titrations

titration is a method of analysing the concentrations
titration is also used to prepare salts

The typical materials needed are:

- 50cm³ burette
- 25cm³ volumetric pipette
- A suitable indicator

the indicator is required to show end-point of titration



Chromatography

• a technique to separate substances that have different solubilities in a solvent.

- Pencil line drawn on paper? Sample placed
- Paper put into solvent container above line
- Solvent travels up paper
- Different substances, travel to different lengths
- Locating agents are substances which react with the sample & produce a coloured product, the agent is used on chromatogram to locate things like amino acids / sugars.

CHROMATOGRAMS

Pure: all substance in same place

Mixture: separate to show all differences

Impure: show more than one spot

$$R_f = \frac{\text{distance travelled by substance}}{\text{distance travelled by solvent}}$$

Retention factor

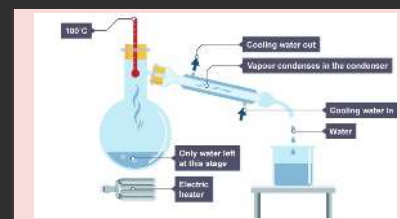
R_f value of a compound is always same.

separation & purification

- **Suitable Solvent**: For mixture of solids, can dissolve some.
- **Filtration**: For undissolved solids from solid & liquid
- **Crystallisation**: Separate dissolved solid by evaporation
- **Simple distillation**: Separate liquid & soluble solid
- **Fractional distillation**: Separate two or more liquids.

Assessing Purity

- pure substances boil & melt at specific temperatures
- mixtures have a range of temperatures
- Melting & boiling point can be used to assess purity
- The closer measured M.P. & B.P. is to the actual point is how pure substance is.



identification of ions and gases

Tests to identify Anions:

Anion	Test	Result
Carbonate (CO ₃ ²⁻)	Add dilute acid and test the gas released	If the carbonate gas produced is CO ₂ which turns lime water milky
Chloride (Cl ⁻)	Acidify with dilute nitric acid and add aqueous silver nitrate	White precipitate formed
Bromide (Br ⁻)	Acidify with dilute nitric acid and add aqueous silver nitrate	Cream precipitate formed
Iodide (I ⁻)	Acidify with dilute nitric acid and add aqueous silver nitrate	Yellow precipitate formed
Nitrate (NO ₃ ⁻)	Add aqueous NaOH and aluminium foil, warm gently and test the gas released	Gas given off is ammonia, has a pungent smell, turns red litmus blue
Sulfate (SO ₄ ²⁻)	Acidify with dilute nitric acid and add aqueous barium nitrate	White precipitate formed
Sulfite (SO ₃ ²⁻)	Add dilute acid, warm gently and test the gas released	Gas decolorises purple acidified aqueous potassium manganate(VII) solution

using NaOH & NH₃ to identify the cations:

NETAL CATION	EFFECT OF ADDING NaOH	EFFECT OF ADDING AMMONIA SOLUTION
ALUMINIUM (Al ³⁺)	WHITE PRECIPITATE FORMED, INSOLUBLE IN EXCESS	WHITE PRECIPITATE FORMED, SOLUBLE IN EXCESS
AMMONIUM (NH ₄ ⁺)	AMMONIA RELEASED (FUMES)	—
CALCIUM (Ca ²⁺)	WHITE PRECIPITATE FORMED, SOLUBLE IN EXCESS	WHITE PRECIPITATE FORMED, INSOLUBLE IN EXCESS
CHROMIUM (Cr ³⁺)	GREEN PRECIPITATE FORMED, SOLUBLE IN EXCESS	GREY-GREEN PRECIPITATE, INSOLUBLE IN EXCESS
COPPER (Cu ²⁺)	PURPLE PRECIPITATE FORMED, INSOLUBLE IN EXCESS	PURPLE PRECIPITATE FORMED, INSOLUBLE IN EXCESS
IRON (III) (Fe ³⁺)	BROWN PRECIPITATE FORMED, INSOLUBLE IN EXCESS	BROWN PRECIPITATE FORMED, INSOLUBLE IN EXCESS
IRON (II) (Fe ²⁺)	GREEN PRECIPITATE FORMED, INSOLUBLE IN EXCESS	GREEN PRECIPITATE FORMED, INSOLUBLE IN EXCESS
ZINC (Zn ²⁺)	WHITE PRECIPITATE FORMED, SOLUBLE IN EXCESS	WHITE PRECIPITATE FORMED, SOLUBLE IN EXCESS

tests to identify gases: Flame test for cations:

Gas	Appearance of gas	Test	Result
Ammonia (NH ₃)	Colourless, pungent smell	Dipped litmus paper	Turns blue
Carbon dioxide (CO ₂)	Colourless and odourless	Bubbles through limewater	Limewater turns milky/cloudy
Chlorine (Cl ₂)	Yellow-green, choking smell	Blue litmus paper	Turns red, solution is bleached
Hydrogen (H ₂)	Colourless and odourless	Hold a lit splint at mouth of test tube	Burns with a 'squeaky pop' sound
Oxygen (O ₂)	Colourless and odourless	Hold a glowing splint	Sparks / relights
Sulfur dioxide (SO ₂)	Colourless, pungent choking smell	Add to acidified aqueous potassium manganate(VII)	Turns from purple to colourless

Cation	Colour of flame
Li ⁺	Red
Na ⁺	Yellow
K ⁺	Lilac
Ca ²⁺	Orange-red
Cu ²⁺	Blue-green
Ba ²⁺	Apple green